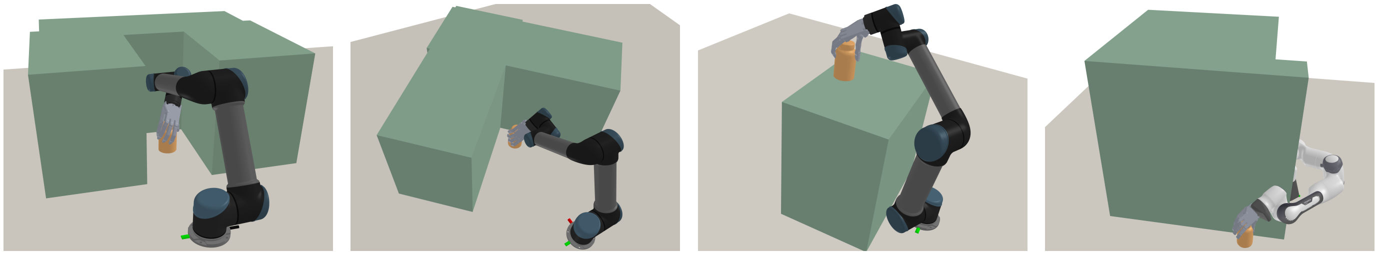

(a) Narrow Corridor (S1) (b) Room Corner (S2) (c) Shelf Top (S3) (d) Partition Board (S4)

Fig. 6: Visualization of four evaluation scenarios (S1-S4) with one successful grasp involving the robotic arm and dexterous hand, the obstacles painted in green, and the grasped object highlighted in orange. Scene S4 is specifically designed to showcase Franka's collision avoidance using its kinematic redundancy.

Appendix A
Additional Results and Analysis

A. Design and Visualization of the Evaluation Scenes

To evaluate the proposed method, we designed six challenging scenarios featuring high obstacle density and grasps near the workspace boundary. Four scenarios (S1–S4) focusing on collision avoidance are visualized in Fig. 6. The remaining two, targeting hand reachability and joint proximity constraints, are illustrated in Fig. 3(b) and (c) in the main text, respectively.

B. Parameter Sensitivity Analysis

We have conducted systematic sensitivity analyzes on key hyperparameters in our method. All experiments are performed on a randomly sampled subset of DexGraspNet (DGN), consisting of 2,308 objects with various geometries, poses, and scales. We analyze the effects of the constraint weight λ_c and the learning rate η_λ under the collision-avoidance constraint in the Narrow Corridor scenario (Fig. 6 (a)). Following common practice in guided diffusion sampling [16], we apply magnitude clipping to the pose gradients to prevent numerical singularities and divergence under large guidance strength. The heatmap results (Fig. 7) reveal consistent trends across the two arms, 1) FR and OSR increase rapidly with larger λ_c and η_λ and quickly enter a saturation regime at moderate-to-high values; 2) GSR exhibits only mild degradation during this process, indicating limited sensitivity to the guidance parameters; and 3) The learning rate affects mainly the convergence speed of OSR rather than the final performance.

The result of the effect of the sigmoid shape (Fig. 8) shows that when the sharpness parameter $\kappa > 50$, all metrics converge and remain stable. This behavior can be attributed to the fact that the signed distance field (SDF) values are typically on the order of $10^{-2} \sim 10^{-1}$ in the real environment, for which κ in the range of $50 \sim 500$ provides an appropriate scale to capture the variations of the SDF, leading to stable optimization.

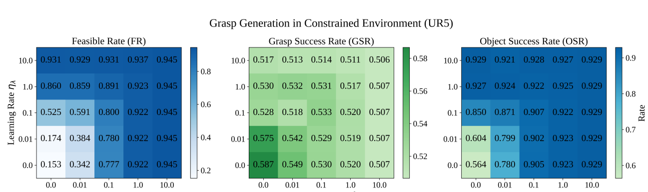

Fig. 7: Effect of different constraint weights λ_c and learning rates η_λ.

In the reachable grasp generation experiment, we analyze the weight coefficient α that balances collision-avoidance and reachability (Fig. 9). Results under a linear sweep of α in [0,1] reveal stable trends across both arms: 1) RFR increases monotonically with α, while CFR exhibits a slight decrease, indicating a clear trade-off between reachability and collision avoidance; 2) FR and GSR demonstrate a balanced relationship, with GSR showing only mild sensitivity to α, reflecting the robustness of the proposed guidance method.

In the grasp generation experiment with reference joint configuration, we analyze the effect of the joint regularization weight λ_s. As shown in Fig. 10, the results reveal a clear trade-off between joint proximity and task success. Specifically, AJP decreases monotonically with increasing λ_s, indicating that the generated grasps remain closer to the reference joint configuration. However, when λ_s exceeds a certain threshold, the grasp gradually deviates from object-centric configurations, leading to a rapid drop in GSR and OSR. This behavior suggests that joint regularization acts as a relatively strong constraint.

We finally evaluate the effect of the number of diffusion denoising steps $\mathrm{T_N}$ during sampling. Experiments are conducted in the Narrow Corridor scenario, with $\mathrm{T_N} = 5, 10, 15, 20$ (Fig.

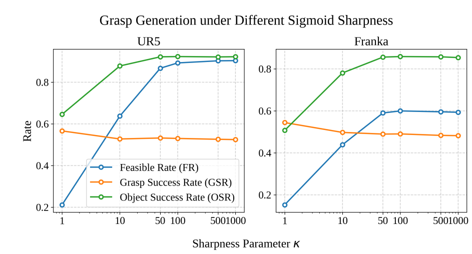

Fig. 8: Effect of different sigmoid shape.

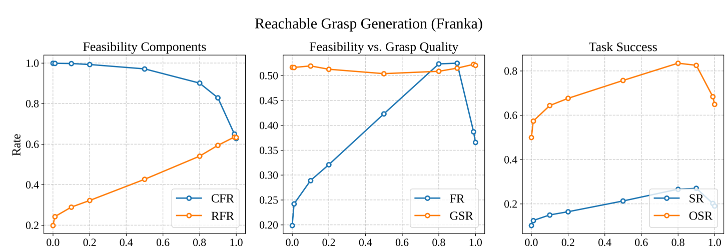

Fig. 9: Effect of different weights coefficient α.

Table VII: Performance of the ablation groups compared to '**Ours**' in Tab. I

Scene	Method	Changes of Metrics(%)↑			
		FR	GSR	SR	OSR
S1	w/ proj.	- 4.80	+ 2.54	- 0.41	+ 0.34
	w/ noised grad.	- 0.75	- 3.69	- 3.64	- 0.12
S2	w/ proj.	- 18.39	+ 5.52	- 5.19	- 1.41
	w/ noised grad.	- 0.12	- 1.96	- 2.29	- 0.11
S3	w/ proj.	+ 0.83	+ 0.81	+ 0.94	+ 0.33
	w/ noised grad.	+ 10.93	- 0.70	+ 5.07	+ 2.48

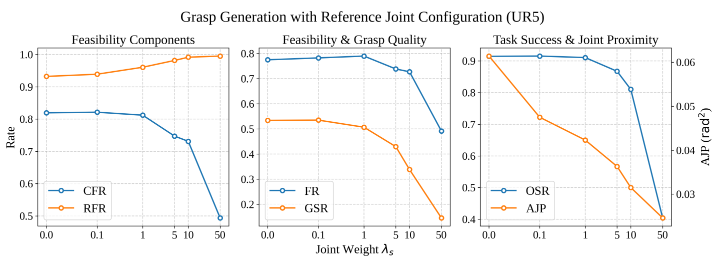

Fig. 10: Effect of different joint weights λ_s.

11). The results show that using a very small number of denoising steps (e.g, 5) leads to noticeable degradation across all metrics. When $\mathrm{T_N} = 10$, OSR already approaches its maximum, indicating that the proposed DDIM-based guided diffusion achieves stable performance with few steps. Further increase $\mathrm{T_N}$ allows for more fine-grained pose refinement, which can improve FR, but may slightly reduce GSR. Considering both sampling efficiency and task success, we adopt $\mathrm{T_N} = 10$ as the default setting in our experiments.

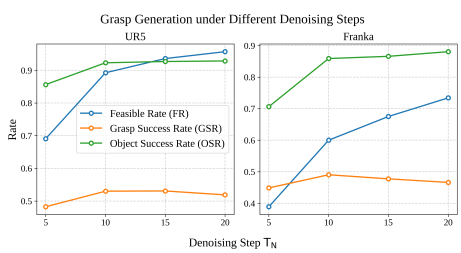

Fig. 11: Effect of different denoising step $\mathrm{T_N}$.

C. Ablation of the Nullspace Projection Method and the Noised Gradient

We propose a nullspace projection method to address the potential decline in grasp quality caused by the introduction of guidance, which ultimately maintains grasp quality. The main task is defined as keeping a point on the palm fixed in the world coordinates. By projecting the guidance gradient ∇g to the main task's nullspace, the hand keeps facing the object in most cases, thus maintaining the grasp quality. The ablation results in Tab. VII show the percentage change of metrics when the nullspace projection is enabled (w/ proj.) compared to '**Ours**' in Tab. I of the main text. As shown by the results, the GSR increases without projection, indicating improved grasp quality. However, there is minimal change in the SR, indicating that the absolute number of successful grasps in a batch remains consistent. This trend arises from the interplay between the decrease in feasible grasps (FR) and the increase in grasp quality (GSR). The decline in FR is expected, as the direction of gradient descent is modified by the projection. The OSR presents similar trends to SR.

When the noised wrist pose $\boldsymbol{x}_t$ instead of $\hat{\boldsymbol{x}}_0$ is used to compute ∇g in (13), we assume that the resulting noised gradient may result in poor convergence. The ablation results with noised gradient (w/ noised grad.) are presented in Tab. VII. The FR decreased by 0.75% and 0.12% in S1 and S2 as expected, while it increased by 10.93% in S3. The obstacles of S1 and S2 result in the robotic arm alternately collide with two walls and produces inconsistent gradients which hinder convergence. While for S3 with less obstacles, $\nabla g(\boldsymbol{x}_t)$ performs better by providing more direct and timely guidance. Therefore, in specific applications, the choice between $\boldsymbol{x}_t$ and $\hat{\boldsymbol{x}}_0$ should be determined by the metrics relevant to the particular scenario. In addition, the grasp quality marginally decreases (GSR), and the change of SR and OSR depends on the specific scene.

D. Computation Time Cost

We report inference-time cost measured as the average inference time per successful grasp. Fig. 12 shows the average inference time evaluated over 2,308 objects, with 10 candidate grasps generated for each object, under different batch sizes on UR5 and Franka. All experiments are conducted on a single NVIDIA RTX 3090 Ti GPU. Across all settings, the proposed method generates one successful grasp within approximately 0.5–2 ms on average. Although gradient-related computations are evaluated at each diffusion step, the inference time per successful grasp remains lower than that of the rejection-based baseline under the same number of denoising steps. This behavior is mainly attributed to the improved feasibility and success rates introduced by guidance, which reduce the number of discarded samples. In addition, inference time decreases with increasing batch size, indicating that the guidance-related computations are efficiently processed in batches.

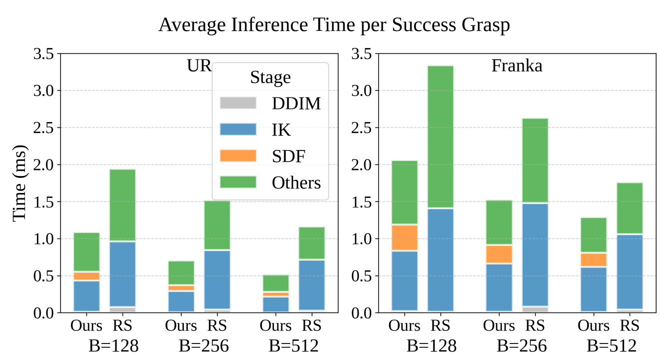

Fig. 12: The average inference time per successful grasp under different batch size on UR5 and Franka in the Narrow Corridor scenario.

The inference time is decomposed into SDF, IK, and other components in Tab. VIII. The SDF query constitutes the dominant portion of the runtime, as it involves evaluating signed distance values for a large number of sampled points on the

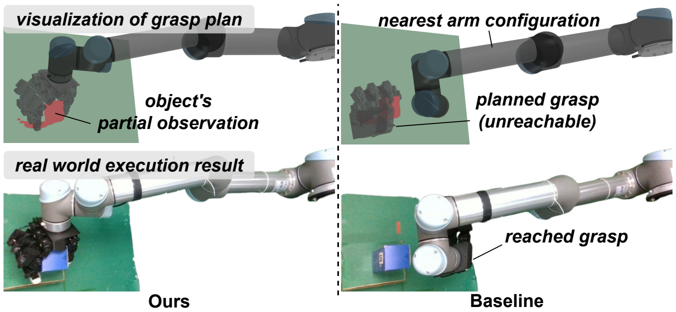

Fig. 13: Comparison of grasp generation near the arm's workspace boundary. The top right figure shows a planned grasp that the robotic arm cannot reach.

robot against each object in the scene. Within the IK component, the computation time is comparably distributed among solver invocation, forward kinematics and Jacobian computation, and error and gradient evaluation, due to repeated calls to the analytical solver and large-scale matrix operations. The remaining computation cost primarily arises from conversions between Lie algebra and Lie group representations.

Table VIII: Inference-Time Breakdown of Major Components

Module	Component	Time Ratio (%)
SDF	Query SDF	56.8
	Link Jacobian	17.8
	Sphere Jacobian	12.2
	Nullspace Projection	6.3
	Others	6.9
IK	Call Solver	41.6
	FK & Jacobian	33.4
	Error & Gradient	24.5
	Others	0.6
Others	Gradient Conversion	57.4
	Point Cloud Encoding	11.5
	Predict Joints	5.2
	Others	25.9

E. Influence of the Number of IK Solutions

Remember that solving (6) involves minimization over the IK solution set $\mathcal{Q}$, which is approximated with a finite set in practice. As shown in Tab. IX, increasing the number of IK solutions enhances all metrics, indicating that a better solution to (6) has been found. Note that using fewer than 4 IK solutions can significantly degrade performance with the UR5, as these solutions reside in disconnected sub-manifolds of the configuration space. Their varying order can result in unstable gradients due to the discontinuity of the IK solutions. In contrast, the performance with Franka is less affected, as it has a connected IK solution set. In addition, more inverse kinematics (IK) solutions reduce the number of grasps generated per second. Thus, an appropriate number—8 for UR5 and 10 for Franka—is chosen.

Appendix B
Additional real-world experiments

A. Generating Reachable Grasps Near Workspace Boundaries

We showcased the effectiveness of the proposed method in generating reachable grasps near the arm's workspace boundary. We executed a generated grasp both with and without guidance, as shown in Fig. 13. Without guidance, the system tends to produce unreachable grasps behind the object, resulting in failures even when the arm is fully extended.

Table IX: Performance of grasp generation with varying number of IK solutions

Arm	IK Num.	Metrics(%)↑			
		FR	GSR	SR	OSR
UR5	2	2.74	44.01	1.21	10.06
	4	45.36	56.64	25.69	83.64
	8	73.16	57.80	42.29	90.32
Franka	5	35.47	48.79	17.31	71.58
	10	46.18	51.78	23.91	80.46
	20	60.70	53.65	32.57	85.72

B. Generating Proximal Grasp Configurations for Regrasping

We demonstrated the proposed method's ability to generate grasps with adjacent arm configurations in joint space during regrasping. Using the initial grasp's arm configuration as a reference, we consecutively sampled grasps 10 times, selecting the one with the lowest joint proximity loss from 40 candidates per sample for execution, as shown in Fig. 14. Without guidance refinement, the baseline method leads to unnecessary arm movements. In contrast, our approach reduces the average joint space distance between grasps from 0.88 rad to 0.56 rad, resulting in a smoother execution trajectory.

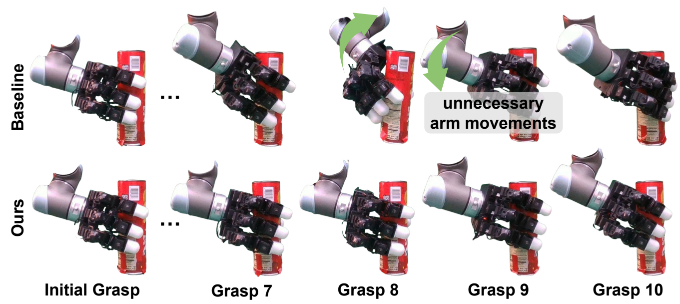

Fig. 14: Illustration of guided grasp generation for continuously regrasping with the Joint Proximity constraint. The reference initial grasp and the four final grasps are displayed from left to right, with additional grasps omitted for brevity. Please refer to the attached video for the complete grasp sequence.

C. Failure Cases of Executing Generated Grasps in the Real World

We present several failure cases in Fig. 15, highlighting potential limitations of the proposed method and possible directions for improvement. 1) In some cases, the guidance compromises the quality of a small subset of generated grasps, resulting in unstable configurations that only grasp the edge of the object (Fig. 15 (a-b)). This accounts for 7 out of 20 failed grasps. This issue often arises when the arm remains in collision until the end of the denoising process, causing conflicts between constraint satisfaction and convergence to the learned grasp distribution. Although we utilize the grasp evaluator to filter out high-quality grasps, we find that real-world point clouds are often noisier and more incomplete, leading to misestimation of grasp success probabilities. This issue can be mitigated through increased data augmentation and fine-tuning with real-world data. Another reason for failure is inaccuracies in motion planning and open-loop execution, which can result in unintended collisions, contributing to 9 out of 20 failed grasps (Fig. 15 (c)). This issue can be mitigated through real-time feedback and online sensing. Additionally, incorrect estimation of the object's mass parameters or insufficient friction accounts for the remaining 4 out of 20 failed grasps (Fig. 15 (d)), though this aspect lies beyond the scope of this work.

Fig. 15: Failure cases in real-world experiments. (a–b) The quality of a small subset of generated grasps is degraded by the guidance, leading to unstable grasps holding only the object's edge. (c) The dexterous hand unintentionally contacts the object due to open-loop execution. (d) The object shifts in hand because of inaccurate mass parameter estimation or insufficient friction.

APPENDIX C
ADDITIONAL DETAILS OF THE PROPOSED METHOD

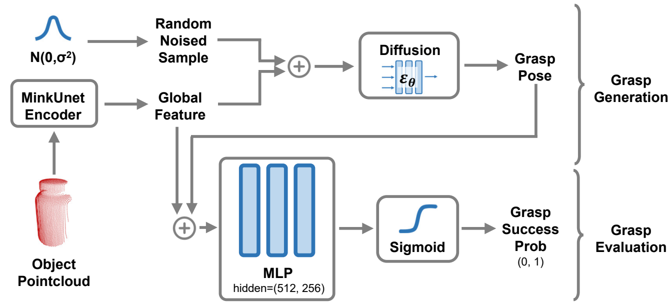

Fig. 16: Architecture of the grasp generation and evaluation networks.

A. *Network Architecture*

Our network design is inspired by prior work [1]. For grasp pose generation, the object's partial point cloud is embedded into a 1024-dimensional global feature vector using a Minkowski Unet. This global feature, along with a randomly sampled noise from a Gaussian distribution, is input into the denoising process. The noise prediction network, conditioned on the global feature and the time step t, predicts a denoised sample from the noisy input. The grasp pose is then recovered from the fully denoised sample and the global feature. A grasp evaluator predicts the probability of successfully executing a given grasp, based on the grasp pose and the global feature. The evaluator is implemented as an MLP with hidden dimensions of $[512, 256]$, followed by a sigmoid activation function. The grasp evaluation uses the same training data as the generation network, augmented with execution success labels from performing all grasps in the MuJoCo simulator. The evaluator is trained using cross-entropy loss and successfully predicts 87% of grasps in the test set with its best checkpoint.